



Chemistry IER Problem-Solving Worksheet

Identify → Execute → Reflect

This worksheet is a mechanical procedure. It eliminates choice. Fill every blank in order. The Trigger Trace tells you which card fires next. The unit trace catches your algebra errors.

Reference: `honors-chemistry-expression-to-equation-vocabulary.md` (213 cards)

Method: `chemistry-without-intuition.md`

Problem ID	Difficulty Level	Domain(s)	Time Spent
	Level 1 / 2 / 3 / 4		_____ min

Level guide: Level 1 = 1–2 cards (direct, one equation). Level 2 = 2–4 cards (one conversion chain). Level 3 = 4–6 cards (two domains, one bridge). Level 4 = 6–10 cards (3+ domain synthesis).

STEP I — IDENTIFY

"Which cards fire? What is given? What is hidden?"

Chemistry Without Intuition, §0.2 — The 4-Pass Extraction

Pass 1 — Five-Word Gist (pencil down)

Read once. No writing. State the core of the problem in five words or fewer.

Pass 2 — Variable Table

Extract every number with its unit. Convert all temperatures to Kelvin immediately if gas laws or thermodynamics are involved (Card 197). Implicit values are not optional — they get their own rows.

#	Variable Name	Symbol	Value	Unit	Source
1					
2					
3					
4					
5					
6					
7					
8					
TARGET			?		

Knowns: ____ | **Unknowns:** ____

Implicit Value Scan

Protocol B from Chemistry Without Intuition, Part VII. Check every box BEFORE moving to Pass 3.

If the problem says...	Then...	Card	Found?	In table?
"STP"	$T = 273 \text{ K}$, $P = 1 \text{ atm}$, $V_m = 22.4 \text{ L/mol}$	7, 186	☐	☐
"strong acid" (by name: HCl, HBr, HI, HNO ₃ , H ₂ SO ₄ , HClO ₄ , HClO ₃)	$[\text{H}^+] = [\text{HA}]_0$ — complete dissociation	107, 189	☐	☐
"strong base" (Group 1 & 2 hydroxides)	$[\text{OH}^-] = [\text{base}]_0 \times (\# \text{ of OH}^-)$	108, 190	☐	☐
"excess" / "in excess"	That reactant is NOT limiting — ignore for yield	188	☐	☐
"standard state" / "°"	298 K, 1 atm, 1 M; ΔH_f° of elements = 0	45, 187	☐	☐
"at equilibrium"	$Q = K$, $\Delta G = 0$, forward rate = reverse rate	192	☐	☐
"first-order"	$t_{1/2} = 0.693/k$ — half-life constant	83, 193	☐	☐
"ideal gas" (stated or implied)	Obeys $PV = nRT$	191	☐	☐
"catalyst present"	E_a lowered; ΔH , ΔG , K unchanged	88, 194	☐	☐
"monoprotic acid"	One ionizable H ⁺ ; 1:1 neutralization	195	☐	☐
"diprotic acid"	Two ionizable H ⁺ ; two equivalence points; $K_{a1} \gg K_{a2}$	196	☐	☐
"water" (specific heat)	$c_w =$	34	☐	☐

If the problem says...	Then...	Card	Found?	In table?
	$4.184 \text{ J g}^{-1} \text{ K}^{-1}$			

Pass 3 — Trigger Trace

Underline every chemistry trigger phrase in the problem statement. For each one, write the card number and the instant equation. The cards choose — you transcribe. The order of cards tells you the calculation chain. Do not rearrange.

Order	Trigger Phrase in Problem	Card #	Instant Equation
1			
2			
3			
4			
5			
6			
7			
8			
9			
10			

Cards fired: ____ | **Domain chain:** _____

Difficulty: 1–2 cards = Level 1. 3–4 = Level 2. 5–6 = Level 3. 7+ = Level 4.

Pass 4 — Unknown

What am I solving for?	Symbol	Unit

Bridge Recognition

Chemistry Without Intuition, Part VI. Two or more distinct chemical steps?

Does this problem chain multiple domains? ☐ Yes ☐ No

If YES — what does the first step produce that the second step needs?

First domain produces...	Bridge variable	Feeds into...	Bridge card
☐ Stoich: n_{gas} from mass	n_{gas}	Gas Law: $V = nRT/P$	160, 161
☐ Stoich + $\Delta H \rightarrow$ heat released	q	Calorimetry: $q = mc\Delta T$	162
☐ ICE table $\rightarrow$ equilibrium amounts	n_{eq} or x	Mass/concentration at equilibrium	163
☐ $\Delta H^\circ, S^\circ \rightarrow \Delta G^\circ$	ΔG°	$\Delta G^\circ = -RT \ln K \rightarrow K$	164
☐ $\Delta G^\circ \rightarrow$ cell potential	ΔG°	$E^\circ = -\Delta G^\circ / (nF)$	165
☐ $\text{p}K_a$ from K_a	$\text{p}K_a$	Henderson-Hasselbalch	169
☐ $E^\circ_{\text{cell}} \rightarrow$ equilibrium	E°	$E^\circ = (RT/nF) \ln K \rightarrow K$	166

Bridge variable: _____ | Second domain: _____

STEP E — EXECUTE

"Apply the card equations in Trigger Trace order. No deviation. The first card's output is the second card's input."

Chemistry Without Intuition, Part I

Card Execution Chain

Step	Card #	Equation Applied	Calculation	Result
1				
2				
3				
4				
5				
6				
7				
8				
9				
10				

The Mole Bridge (stoichiometry problems)

Chemistry Without Intuition, Part VII — Protocol A. If stuck: convert everything to moles.

```

grams —[÷ M]→ moles —[mole ratio]→ moles —[× M]→ grams
      |
      |—[× 22.4 @ STP]→ L gas
      |—[× 6.022×1023]→ particles
      |—[÷ V (L)]→ molarity (M)

```

Mole bridge path for this problem:

_____ → _____ → _____ → _____ → **TARGET**

Unit Trace

Trace the units from the first given quantity through every operation. If they do not cancel to the target unit, the card order is wrong or a conversion was missed. Chemistry errors almost always betray themselves through units.

_____ → _____ → _____ → _____ → _____ → **TARGET**

Units cancel correctly? ☐ Yes ☐ No → Check card order

Final Calculation

(Write the complete numerical calculation here)

Answer

Value: _____ **Unit:** _____

Significant figures: Given: _____ sf → Answer: _____ sf | Match? ☐ Yes ☐ No

Boxed answer: _____

STEP R — REFLECT

"Three mechanical checks. Pass all three before moving to the next problem."

Check 1: Unit Cancellation

Did the units trace confirm correct cancellation? ☐ Yes ☐ No

Check 2: Significant Figures

Least precise given value	SF	Your answer's SF	Match?
			☐

Check 3: Physical Plausibility

Sanity check	Expected range	Your answer	Pass?
% yield $\leq 100\%$?	0–100%		☐
pH in 0–14? (25°C)	0–14		☐
Mass positive?	> 0		☐
K value consistent with ΔG° sign?	$K > 1$ if $\Delta G^\circ < 0$		☐
Temperature in Kelvin positive?	> 0		☐

Error Log (complete only if answer was wrong)

"The error log is a vaccine. Identify the broken link in the chain. Fix it. Prevent it."

Error type(s):

Category	Specific detail (which variable? which step? which card?)	[]
Skipped mole conversion (g $\rightarrow$ mol before stoich)		[]
Balancing coefficient error		[]
Limiting reagent error (did not divide mol by coefficient)		[]
Unit mismatch — K vs $^{\circ}\text{C}$		[]
Unit mismatch — J vs kJ		[]
Unit mismatch — atm vs torr vs Pa		[]
Unit mismatch — L vs mL		[]
ICE table setup error (sign or stoichiometry in Change row)		[]
Small- K approximation not verified (5% rule, Card 94)		[]
Le Chatelier direction reversed		[]
Wrong R value (0.08206 vs 8.314, Card 20)		[]
Sig fig error		[]
Implicit value missed (STP, strong acid, K_w , etc.)		[]

Category	Specific detail (which variable? which step? which card?)	[]
Δn counted solids/liquids (K_p - K_c , Card 90)		[]
Molar mass not converted to kg/mol for v_{rms} (Card 30)		[]
Henderson-Hasselbalch used outside valid ratio range (Card 112)		[]
Equivalence point pH assumed = 7 for weak acid/base titration		[]
van't Hoff factor i omitted (colligative properties, Card 144)		[]
ΔS (J) not converted before Gibbs: $\Delta G = \Delta H - T\Delta S$ (Card 48)		[]
Other:		[]

Root cause (one sentence):

Prevention (actionable, specific):

Post-Solve Summary

Item	Value
Cards fired	
Domains crossed	

Item	Value
Domain list (in order)	
Pivot card(s)	
ICE table used?	☐ Yes ☐ No
Quadratic required?	☐ Yes ☐ No
Approximation used?	☐ Yes (5% check: _____% → ☐ Pass ☐ Fail) ☐ No
Mole bridge used?	☐ Yes ☐ No
Architecture	☐ Single-domain ☐ Two-domain bridge ☐ 3+ domain synthesis

QUICK-REFERENCE PANEL

Keep this panel available while solving. Contains the most commonly needed data from the 213-card vocabulary deck.

Essential Card Groups

Domain	Cards	Core Equation
Stoichiometry	1–14	$n = m/M$, mole ratio, limiting reagent (smallest n/coeff)
Gases	19–25	$PV = nRT$ ($R = 0.08206$ for atm·L; $R = 8.314$ for J)
Thermochemistry	33–52	$q = mc\Delta T$, $\Delta H = \sum \Delta H_f^\circ(\text{P}) - \sum \Delta H_f^\circ(\text{R})$
Kinetics	76–88	Rate = $k[A]^m[B]^n$, first-order $t_{1/2} = 0.693/k$
Equilibrium	89–100	$K_c = [C]^c[D]^d/([A]^a[B]^b)$, ICE table
Acids & Bases	101–120	pH = $-\log[\text{H}^+]$, Henderson-Hasselbalch

Domain	Cards	Core Equation
Electrochemistry	121–132	$E_{\text{cell}}^{\circ} = E_{\text{cathode}}^{\circ} - E_{\text{anode}}^{\circ}$, Nernst
Solutions	133–144	$M = n/V$, $\Delta T_b = iK_b m$, $\Pi = iMRT$
Nuclear	145–155	$N = N_0 e^{-\lambda t}$, $t_{1/2} = 0.693/\lambda$
Bridges	160–170	Stoich→Gas, Thermo→Eq, $K \rightarrow E^{\circ}$

Implicit Value Catalog (Cards 186–196)

Trigger	Hidden Value	Card
"STP"	$T = 273 \text{ K}$, $P = 1 \text{ atm}$, $V_m = 22.4 \text{ L/mol}$	7, 186
"standard state" / "°"	298 K, 1 atm, 1 M; ΔH_f° of elements = 0	45, 187
"excess"	Reactant is NOT limiting	188
"strong acid" (HCl, HBr, HI, HNO ₃ , H ₂ SO ₄ , HClO ₄ , HClO ₃)	100% dissociation; $[\text{H}^+] = [\text{HA}]_0$	107, 189
"strong base" (Group 1 & 2 hydroxides)	100% dissociation	108, 190
"ideal gas"	Obeys $PV = nRT$	191
"at equilibrium"	$Q = K$, $\Delta G = 0$, forward rate = reverse rate	192
"first-order"	$t_{1/2} = 0.693/k$ — half-life constant	83, 193
"catalyst present"	E_a lowered; ΔH , ΔG , K unchanged	88, 194

Trigger	Hidden Value	Card
"monoprotic acid"	1 ionizable H ⁺ ; 1:1 with base	195
"diprotic acid"	2 ionizable H ⁺ ; $K_{a1} \gg K_{a2}$; two equivalence points	196

Numerical Constants

Constant	Symbol	Value
Avogadro's number	N_A	$6.022 \times 10^{23} \text{ mol}^{-1}$
Gas constant (L·atm)	R	$0.08206 \text{ L atm mol}^{-1} \text{ K}^{-1}$
Gas constant (J)	R	$8.314 \text{ J mol}^{-1} \text{ K}^{-1}$
Faraday constant	F	$96,485 \text{ C/mol e}^{-}$
Specific heat of water	c_w	$4.184 \text{ J g}^{-1} \text{ K}^{-1}$
K_w at 25°C	K_w	1.0×10^{-14}
pH + pOH at 25°C	—	= 14.00
STP molar volume	V_m	22.4 L/mol
1 atm	—	= 760 torr = 760 mmHg = 101.325 kPa
°C → K	—	$T(\text{K}) = T(^{\circ}\text{C}) + 273.15$

The 12 Most Common Pitfalls (Cards 20–144)

#	Pitfall	Consequence	Card
1	T not in Kelvin	Gas law, thermo, Arrhenius	197

#	Pitfall	Consequence	Card
		all break	
2	ΔH (kJ) vs ΔS (J) mismatch in $\Delta G = \Delta H - T\Delta S$	ΔG off by $\times 1000$	48
3	Wrong R for pressure unit	$PV = nRT$ gives nonsense	20
4	Solids/liquids in K expression	K value wrong	89
5	Small- K approx without 5% verification	May need quadratic	94
6	Molar mass in g/mol for $v_{\text{rms}} = \sqrt{3RT/M}$	v_{rms} wrong by $\sim \times 32$	30
7	Δn in $K_p = K_c(RT)^{\Delta n}$ counts all species	K_p wrong	90
8	Only T changes K — forgetting this in Le Chatelier	Misapplied shift direction	99
9	Henderson-Hasselbalch outside valid ratio (0.1–10)	pH unreliable	112
10	Equivalence point pH = 7 for weak acid/base	pH off by several units	115, 116
11	van't Hoff i omitted for ionic solutes	Colligative properties too small	144
12	Oxidation number of O = -2 in peroxides / with F	Redox balancing wrong	122

Problem-Type Fast Routing

Problem opens with...	Domain	Core Cards
"How many grams..." / "What mass..." / "limiting" / "yield"	Stoichiometry	1, 8, 9, 10, 11, 12, 13
"A gas occupies..." / "pressure" / "volume" / "STP"	Gases	19, 20, 21, 26, 27
"How much heat..." / " ΔH " / "calorimeter" / "Hess"	Thermochemistry	33, 34, 39, 42, 43, 48
"Rate..." / "half-life" / "order" / "activation energy"	Kinetics	76, 78, 80–85
"At equilibrium..." / " K_c " / " K_p " / "shift"	Equilibrium	89, 90, 91, 93, 94, 96, 99
"pH of..." / " $[H^+]$ " / "buffer" / "titration" / "indicator"	Acids & Bases	101–120
"Cell potential..." / " E° " / "electrolysis" / "redox"	Electrochemistry	121–132
"Boiling point..." / "freezing point..." / "osmotic pressure"	Solutions	134, 140–144
"Half-life..." / "decay" / "alpha" / "beta" / "binding energy"	Nuclear	145–155

"The exam is won before you enter the room. Two hundred thirteen cards have trained one reflex: see a phrase, write the equation. This worksheet trains a second: chain those equations mechanically, verify with unit analysis, and never guess."

— *Chemistry Without Intuition*

First 20 problems: Fill every section. It will feel slow — you are building the protocol.

Problems 20–50: The cycle accelerates. The Trigger Trace will fire cards before you consciously read the phrase.

Problems 50+: The worksheet becomes a mental checklist. You will catch unit mismatches, missing implicit values, and sign errors before you write them.

Difficulty = number of cards fired. Level 1 fires 2 cards. Level 4 fires 8. The method is identical. Count the cards. Trace the triggers. Execute.